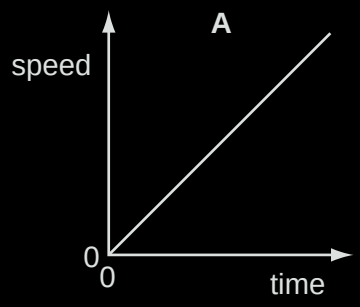
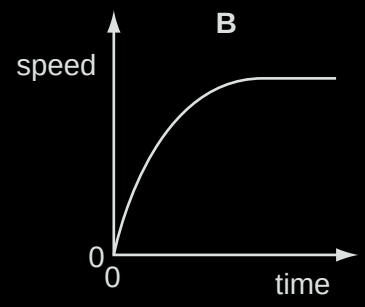
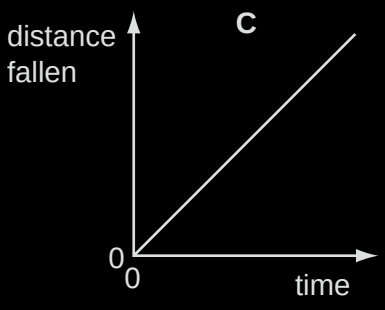
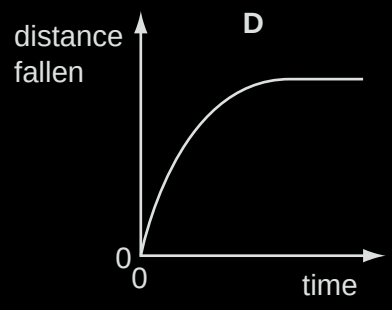








2 A force of 20 N pushes an object of mass 5.0 kg along a rough horizontal surface where the frictional force is 5.0 N.

What is the acceleration of the object?

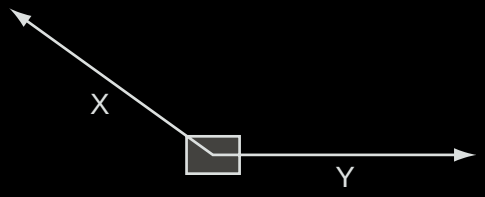
A $1.0\,m/s^2$ **B** $2.0\,m/s^2$ **C** $3.0\,m/s^2$ **D** $4.0\,m/s^2$

3 The diagram shows a vernier scale.

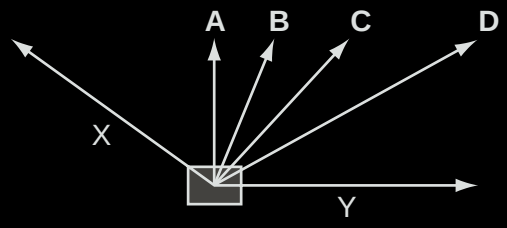


What is the reading on the vernier scale?

A 6.50 cm **B** 6.55 cm **C** 7.00 cm **D** 7.05 cm

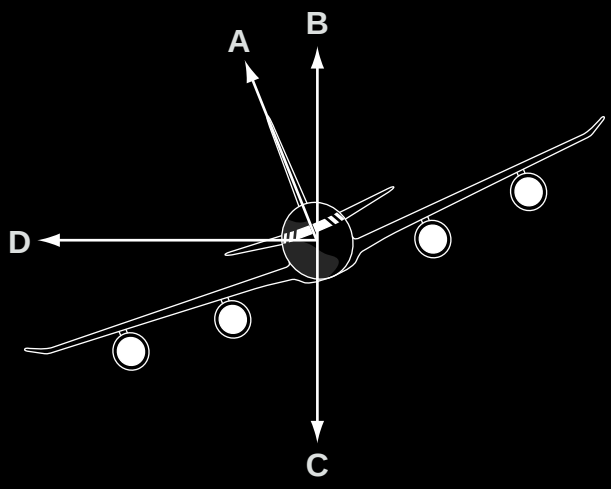


In which direction is the resultant force acting?



5 The diagram shows an aeroplane turning in a horizontal circle at constant speed.

In which direction is there a resultant force?



6 The inertia of a body is its resistance to changes in motion.

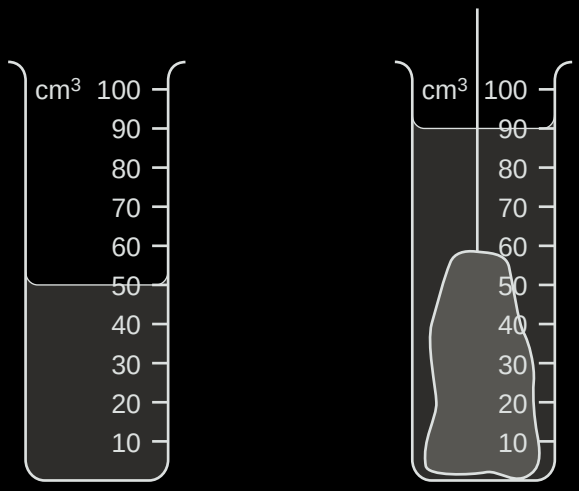
Which property is a measure of the body's inertia?

A its density

B its mass

C the height of its sides

D the size of its base



What is the density of the material from which the object is made?

A $0.4\,g/cm^3$

B $0.9\,g/cm^3$

C $1.1\,g/cm^3$

D $2.5\,g/cm^3$

8 If a nut and bolt are difficult to undo, it may be easier to turn the nut by using a longer spanner.

This is because the longer spanner gives

A a larger turning moment.

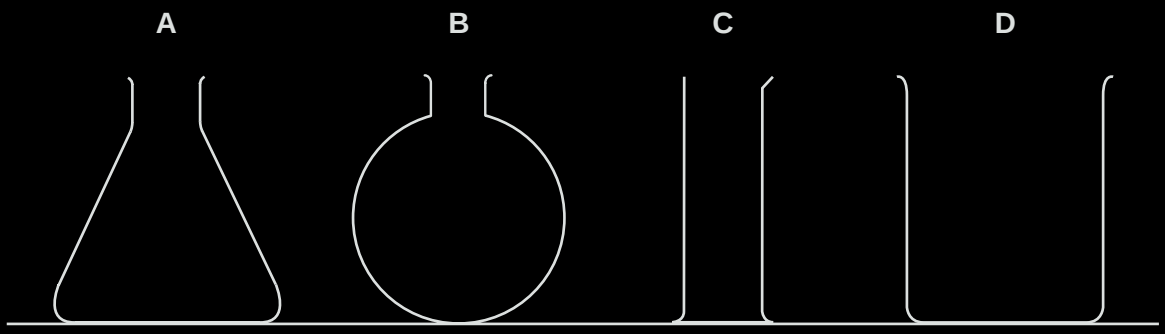
B a smaller turning moment.

C less friction.

D more friction.

9 Some containers are made from thin glass.

Which empty container is the **most** stable?





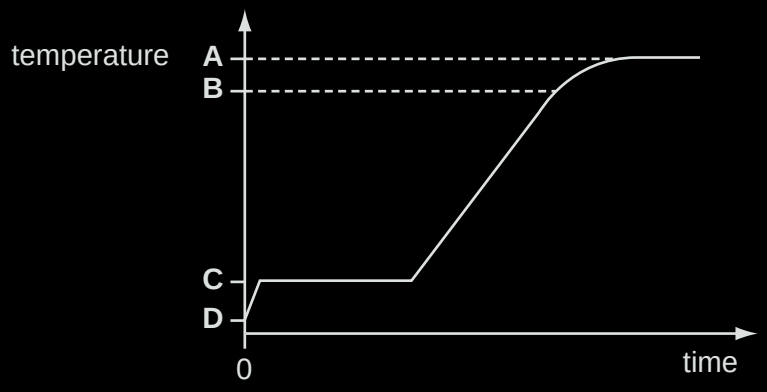
Which of the following statements is correct?

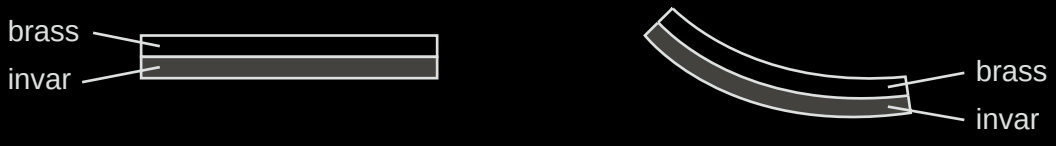
A The water exerts the greatest pressure on the base of vessel **P**.

B The water exerts the greatest pressure on the base of vessel **S**.

C The water exerts the same force on the base of each vessel.

D The water exerts the same pressure on the base of each vessel.


temperature
A
B
C
D
0
time



at room temperature below room temperature

The change in shape occurs because

A brass contracts more than invar.

B brass expands when it cools down.

C invar and brass contract by equal amounts.

D invar contracts more than brass.

20 Density changes are responsible for which method of thermal energy transfer?

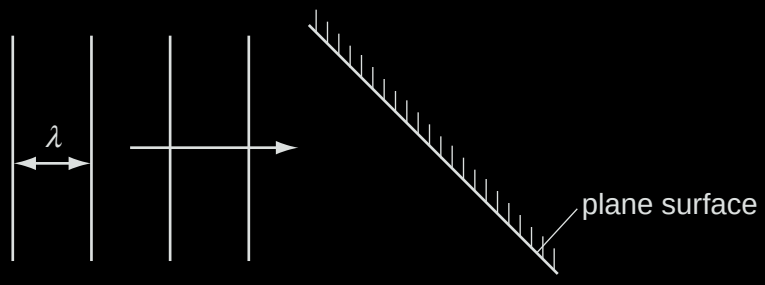
A conduction only

B convection only

C radiation only

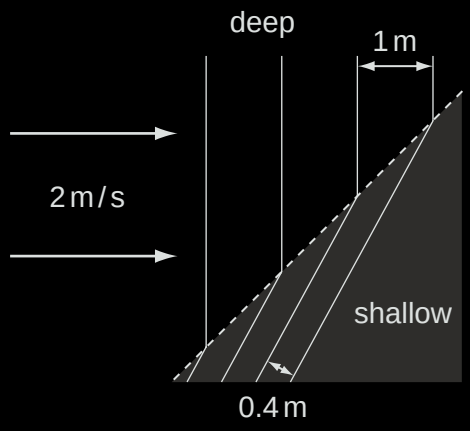
D conduction, convection and radiation

21 In an experiment using a ripple tank, plane wavefronts arrive at a plane surface.



Which of the following correctly describes the waves after they are reflected from the surface?

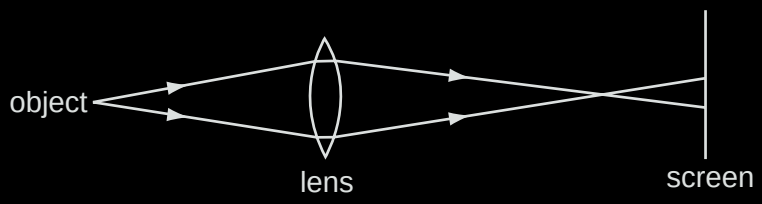
	speed of waves	wavelength λ
A	faster	shorter
B	slower	longer
C	slower	shorter
D	the same	the same



What is the speed of the waves in the shallow water?

A 0.2 m / s **B** 0.8 m / s **C** 2.0 m / s **D** 5.0 m / s

23 A lens forms a blurred image of an object on a screen.



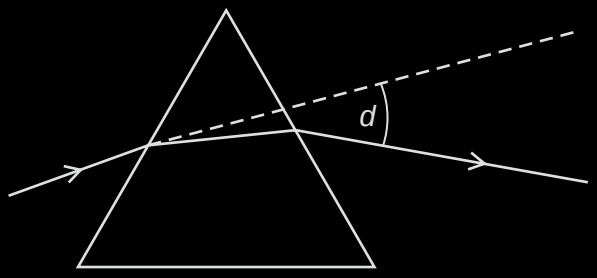
How can the image be focussed on the screen?

A by moving the object away from the lens and screen

B by moving the screen away from the lens and object

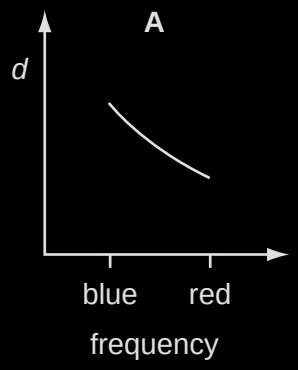
C by using a brighter object at the same position

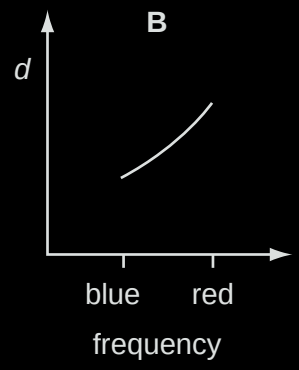
D by using a lens of longer focal length at the same position

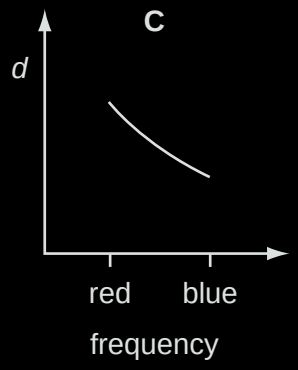


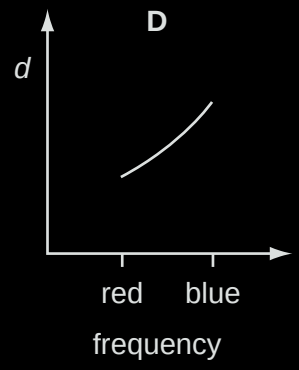
The deviation angle *d* is measured for light rays of different frequency, including blue light and red light.

Which graph is correct?



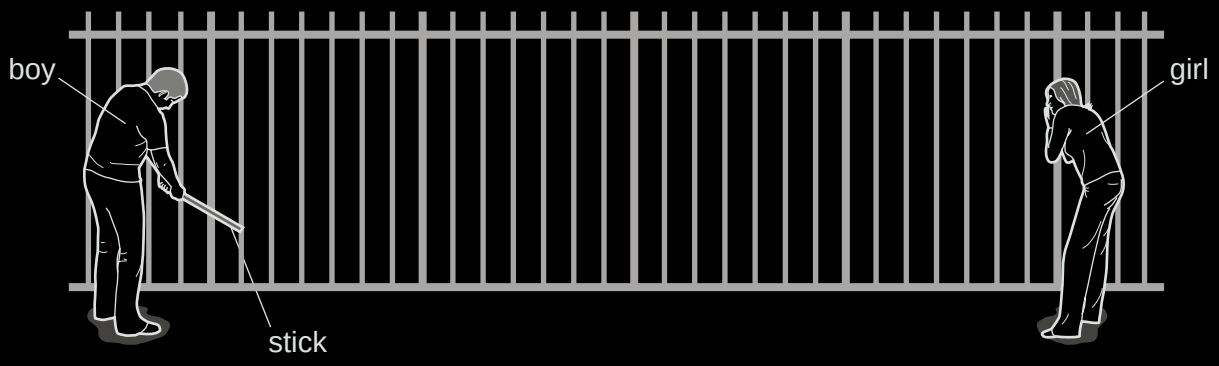






25 Which wave is part of the electromagnetic spectrum?

	$\frac{\text{speed}}{\text{m/s}}$	type
A	330	longitudinal
B	330	transverse
C	3×10^8	longitudinal
D	3×10^8	transverse



How long will it take for the sound to reach the girl through the fence?

A 0 second

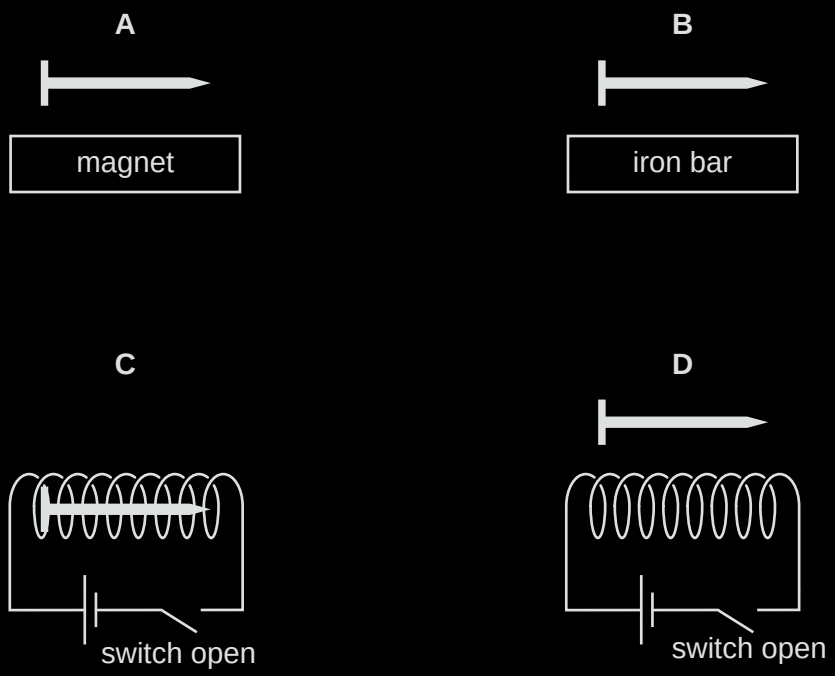
B less than 1 second

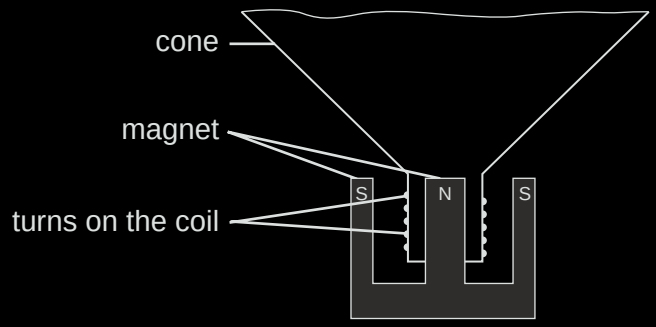
C 1 second

D more than 1 second

27 The diagrams show an iron nail in four different situations.

In which diagram will the nail become an induced magnet?





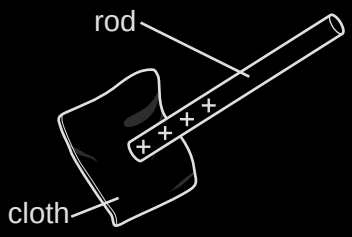
Which type of current is passed through the coil and why?

	current passed through coil	reason why
A	alternating	to keep the magnetic field constant
B	alternating	to make the coil vibrate
C	direct	to keep the magnetic field constant
D	direct	to make the coil vibrate

29 In an electrostatics experiment, a plastic rod is rubbed with a cloth. The cloth becomes negatively charged.

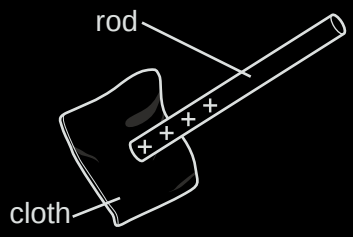
Which diagram shows the charge on the rod, and describes the movement of charge?

A



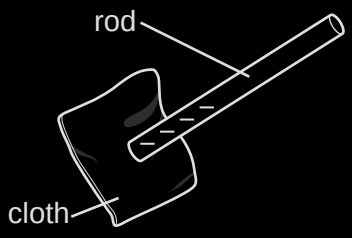
electrons move from the rod onto the cloth

B



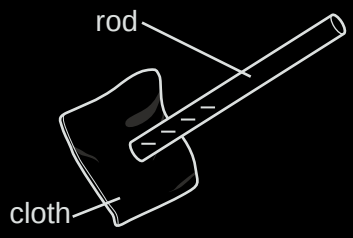
protons move from the cloth onto the rod

C

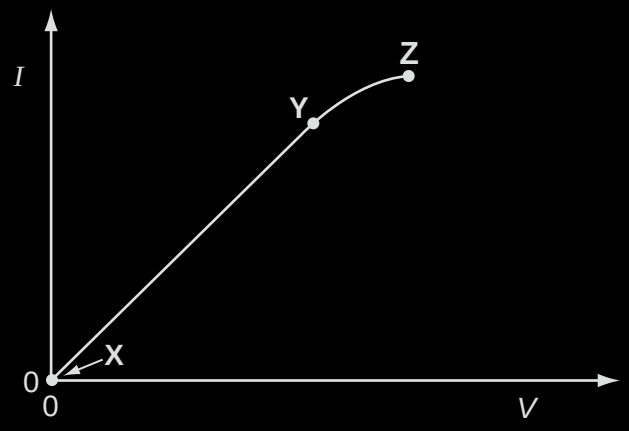


electrons move from the cloth onto the rod

D



protons move from the rod onto the cloth



Where can Ohm's law be applied to the wire?

A at **Y** only

B at **Z** only

C from **X** to **Y**

D from **X** to **Z**

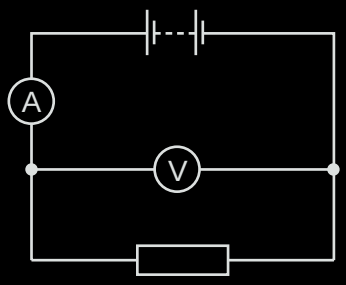


diagram 1

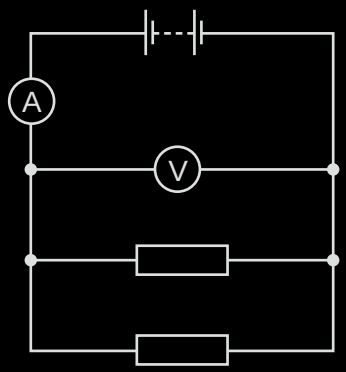


diagram 2

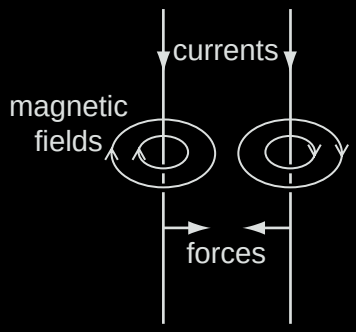
What are the ammeter and voltmeter readings in the circuit shown in diagram 2?

	ammeter reading / A	voltmeter reading / V
A	1.0	3.0
B	1.0	1.5
C	0.5	6.0
D	0.5	3.0

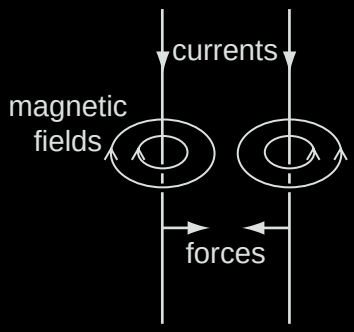
33 How much energy is converted in a resistor of 5.0 Ω carrying a current of 2.0 A for 10 seconds?

A 4.0 J **B** 25 J **C** 100 J **D** 200 J

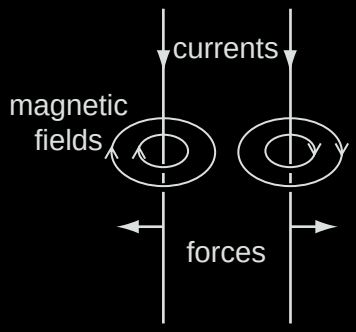
A


currents
magnetic fields
forces

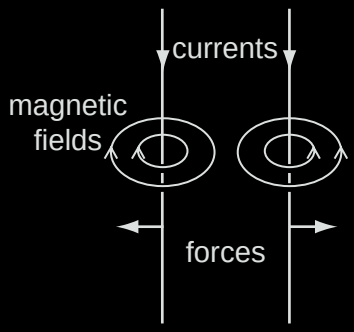
B

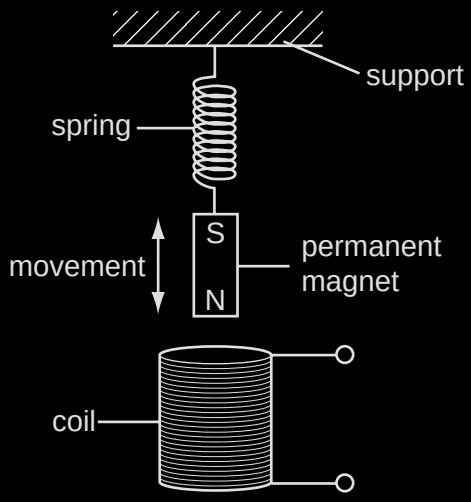

currents
magnetic fields
forces

C


currents
magnetic fields
forces

D


currents
magnetic fields
forces



Which factor, on its own, would **decrease** the maximum value of the induced e.m.f.?

A increasing the number of turns in the coil

B increasing the strength of the magnet

C raising the coil

D raising the support of the spring

36 Why is a transformer used to connect a generator in a power station to a long distance transmission line?

A to decrease the voltage and decrease the current

B to decrease the voltage and increase the current

C to increase the voltage and decrease the current

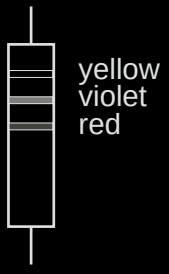
D to increase the voltage and increase the current

37 The table shows part of the colour code for resistors.

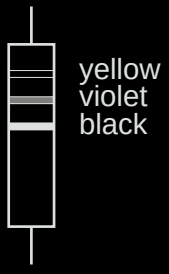
black	brown	red	orange	yellow	green	blue	violet
0	1	2	3	4	5	6	7

Which resistor has a value of 4700 Ω?

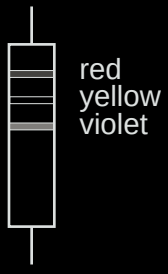
A



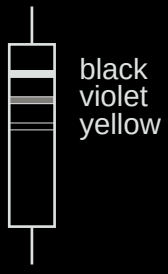
B

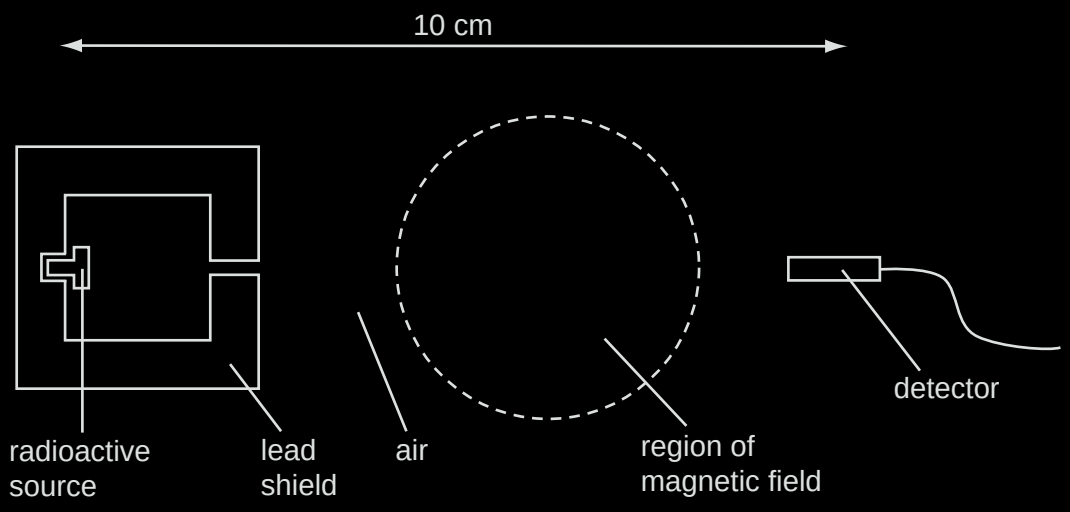


C



D





The results are shown.

	average count per minute
without magnetic field	4500
with magnetic field	2000
background radiation	50

Which radioactive source produced these results?

source	emissions from source
A	alpha-particles and gamma-rays only
B	beta-particles only
C	beta-particles and gamma-rays only
D	gamma-rays only

39 A detector is used to measure the count-rate near a radioactive source. The reading is 4000 counts per minute. After 30 minutes the count-rate has fallen to 500 counts per minute.

What is the half-life of the radioactive source? You may ignore the effects of background radiation.

A 3 minutes

B 5 minutes

C 6 minutes

D 10 minutes